Week 3 Discussion

Covers: conditional probability, Bayes' rule, independence and random variables. Problems 1 and 3 are adapted from C&B §1.7.

Problem 1 (Noisy telegraph: Bayes' rule and independence; C&B Exercise 1.41)

As in Example 1.3.6, telegraph signals "dot" and "dash" are sent in the proportion $3:4$. Erratic transmissions cause a dot to become a dash with probability $\frac{1}{4}$ and a dash to become a dot with probability $\frac{1}{3}$.

Useful tools: the law of total probability, $P(B) = \sum_i P(B \mid A_i)P(A_i)$, and Bayes' rule (Theorem 1.3.5), $P(A_i \mid B) = P(B \mid A_i)P(A_i)/P(B)$.

(a) Write down $P(\text{dot sent})$, $P(\text{dash sent})$, and the four conditional probabilities $P(\text{received} \mid \text{sent})$.

(b) Find $P(\text{dash received})$. Which partition of S are you using?

(c) If a dash is received, what is the probability that a dash was sent?

(d) Now suppose a two-symbol message is sent. Assume the two symbols are chosen independently, each with the $3:4$ proportions, and that transmission errors happen independently for the two symbols. If the message dot-dot is received, what is the probability distribution of the four possible messages (dot-dot, dot-dash, dash-dot, dash-dash) that could have been sent?

Hint for (d): first find $P(\text{dot sent} \mid \text{dot received})$ and $P(\text{dash sent} \mid \text{dot received})$ for a single symbol. Then explain why independence lets you multiply.

▶ *Show answer*

Problem 2 (Pairwise vs. mutual independence)

Recall Definition 1.3.12: $A_1, \dots, A_n$ are **mutually independent** if the product rule holds for *every* subcollection, not just for pairs.

(a) Roll two fair dice, with all 36 ordered outcomes equally likely. Let A = "the first die is odd," B = "the second die is odd," and C = "the sum is odd." Find $P(A)$, $P(B)$, $P(C)$, and check whether each of the pairs (A, B), (A, C), (B, C) is independent.

(b) Are A, B, C mutually independent? Compute $P(A \cap B \cap C)$ and $P(C \mid A \cap B)$, and explain in words why learning A alone tells you nothing about C, while learning A and B together tells you

everything.

(c) How many equations must be checked to show that four events A_1, A_2, A_3, A_4 are mutually independent? List them by size of the subcollection.

(d) A system has three components. Components 1 and 2 are in **parallel** (the pair works if at least one of them works), and that pair is in **series** with component 3 (the system works only if the pair works *and* component 3 works). Let A_i = "component i works," with A_1, A_2, A_3 mutually independent and $P(A_i) = p$.

1. Write the event "the system works" in terms of A_1, A_2, A_3.
2. Show that $P\big(A_3 \cap (A_1 \cup A_2)\big) = P(A_3)\,P(A_1 \cup A_2)$. *Hint:* distribute, then use Theorem 1.2.9(b). Which of your equations needs the **triple** product rule, so that pairwise independence would not be enough?
3. Find $P(\text{system works})$ as a function of p, and evaluate it at $p = 0.9$.

▶ *Show answer*

Problem 3 (Guessing on a standardized test: a random variable; C&B Exercise 1.44)

A test has 20 multiple-choice questions, each with 4 possible answers. A student guesses on every question, so the taking of the exam can be modeled as 20 independent events, each "correct" with probability $\frac{1}{4}$.

(a) Describe the sample space S. How many outcomes does it have? Are they equally likely? *Hint:* record each answer as correct (C) or wrong (W).

(b) Let X = the number of questions answered correctly. Explain why X is a random variable in the sense of Definition 1.4.1. What is its range $\mathcal{X}$?

(c) Fix one outcome s with exactly k correct answers, for example $s = \text{CC}\cdots\text{C}\,\text{WW}\cdots\text{W}$. Use independence to find $P(\{s\})$.

(d) Use the induced probability function (1.4.1), $P(X = k) = P(\{s \in S : X(s) = k\})$, together with counting, to show that

$$P(X = k) = \binom{20}{k}\left(\frac{1}{4}\right)^{k}\left(\frac{3}{4}\right)^{20-k}, \qquad k = 0, 1, \ldots, 20.$$

(e) Find the probability that the student gets at least 10 questions correct. Also find $P(X \geq 1)$ using the complement.

▶ *Show answer*